

PERFORMANCE WORK STATEMENT (PWS)
FOR
Demilitarization Environmental Testing

1.0 General:

1.1 Scope: The contractor shall provide all personnel, equipment, tools, materials, supervision, quality control and other items and non-personal services necessary to perform water and soil testing, on-site observations, and process audits to yield complete environmental testing as defined in this PWS to ensure compliance with Resource Conservation and Recovery Act (RCRA) Subpart X; Permit No. PA6213820503, Application/Permit/Site (APS) No. 960338, Authorization No. 1215612, OnBase Reference ID 1730 and specifically Subpart X, Environmental Sampling Plans, Appendix E-1 and Appendix E-2. The contractor shall perform to the standards in this contract ensuring RCRA sampling and testing compliance in accordance with Pennsylvania Department of Environmental Protection (PADEP) approved sampling plans.

1.2 Background: Letterkenny Munitions Center (LEMC) has an active demolition site located in Chambersburg Pennsylvania where munitions and explosives are routinely disposed of through controlled detonations and confined disposal methods. These activities are fully permitted, RCRA Subpart X; Permit No. PA6213820503, APS No. 960338, Authorization No. 1215612, OnBase Reference ID 1730. Activities include open detonation, open burning, and closed disposal methods of a variety of explosive materials. As a result of these operations, the site is considered a potential source of environmental contamination. Repeated demolition events can introduce hazardous constituents into the surrounding soil and surface/groundwater systems. In addition, to direct contamination, from detonation residues, there is concern regarding stormwater runoff transporting contaminants locally away from permitted sites into nearby surface water bodies and groundwater aquifers. The purpose of this environmental sampling program is to document that activities conducted within the Open Burn/Open Detonation (OB/OD) areas are not adversely impacting the surrounding environment. To date, PADEP has never required us to clean up contaminated soil or sediment from the OB/OD Grounds. Due to the specialized nature of the site, the contractor must be familiar with explosive contaminated environments.

1.2.2 Safety Laws/Codes: The contractor shall comply with all Federal, State and local environmental and safety laws and shall be responsible for all environmental actions and coordination applicable to Federal, State, and local agencies as may be required to execute their portion of this contract. This shall include, but not be limited to, all assessments, studies, permitting and coordination required to comply with Federal, state and local laws regarding endangered species, historic/archaeological sites, and hazardous/toxic materials. The Government will perform assessments necessary to determine and define existing environmental conditions within the boundaries of the Government reservation at the start of the performance of this contract.

1.3 Period of Performance (PoP): The Period of Performance shall be one (1) Base ordering period of five (5) years. The Period of Performance reads as follows:

1.3.1 BASE Ordering Period of Five Years: 15 June 2026 through 14 June 2031

1.4 General Information:

1.4.1 Quality Assurance (QA): The Government shall evaluate the contractor's performance under this contract in accordance with the areas identified in Technical Exhibit 1, Performance Requirements Summary. This plan is primarily focused on what the Government must do to ensure that the contractor has performed in accordance with the performance standards. It defines how the performance standards will be applied, the frequency of surveillance, and acceptable quality level(s) (defect rate(s)).

1.4.2 Recognized Holidays: The following link provides all recognized US holidays: <https://www.opm.gov/policy-data-oversight/pay-leave/federal-holidays/>. The contractor shall not, unless stated as an emergency request by the Government, perform services on these days.

1.4.2.1 New Year's Day

1.4.2.2 Martin Luther King, Jr.'s Birthday

1.4.2.3 President's Day

1.4.2.4 Memorial Day

1.4.2.5 Juneteenth

1.4.2.6 Independence Day

1.4.2.7 Labor Day

1.4.2.8 Columbus Day

1.4.2.9 Veteran's Day

1.4.2.10 Thanksgiving Day

1.4.2.11 Christmas Day

1.4.2.12 When an observed holiday falls on a Tuesday or Thursday, the Depot regularly schedules a closure on the corresponding Monday or Friday.

1.4.3 Place and Performance of Service: The contractor shall provide testing and analysis between the hours of 0700 and 1600, on Monday through Thursday, and Friday with Technical Point of Contact (TPOC) authorization, except on recognized US holidays or when the Government facility/installation is closed due to local or national emergencies, administrative closings, or similar Government-directed facility/installation closings. Performance shall be at LEMC Demil grounds and Building 8001, and the specific location will be provided by the TPOC. The contractor shall always maintain an adequate workforce for the uninterrupted performance of all tasks defined within this PWS when the Government facility/installation is not closed for the above reasons. When hiring personnel, the contractor shall keep in mind that the stability and continuity of the workforce are essential.

1.4.4 Contract Manager (CM): The contractor shall provide a CM who shall ensure performance under this contract. The name of this person, and an alternate who shall act for the contractor when the CM is absent, shall be designated in writing to the KO. The CM or alternate shall have full authority to act for the contractor on all contract matters relating to the daily operation of this contract.

1.4.5 Environmental Compliance: Contractor shall comply with all applicable federal, state, and local environmental laws, statutes, regulations, executive orders, permits, Army regulations (with supplements), as well as Major Subordinate Command (MSC) and installation regulations and policies. The contractor shall immediately report any conflicts between applicable federal, state, local environmental laws, statutes, executive orders, and provisions of Army Regulation 200-1, and any specifications within this contract to the TPOC, as well as the Directorate of Public Works, Environmental Management Division.

1.4.6 Compliance with License and Certification: Compliance with License and Certification Requirements: Contractor shall obtain all license and certification required by federal, state, and local environmental laws and regulations necessary to adhere to the specifications of this contract. The Contractor shall submit all plans, notifications, reports, submittal documents, and fees required by federal, state, and local environmental laws and regulations to the appropriate federal, state, and local authority and/or agency as necessary to adhere to the specification of this contract. All required licenses and certifications required by federal, state, and local environmental laws and/or regulations shall be considered a contract deliverable upon award. The Contractor is responsible for certification of all manifests and total management of their testing and analyzing procedures including scheduling, control, and reporting.

1.4.6.1 Threat Awareness Reporting Program For all contractors with security clearances. Per AR 381-12 Threat Awareness and Reporting Program (TARP), contractor employees must receive annual TARP training by a CI agent or other trainer.

1.4.6.2. Per PADEP, Professional Geologist (PG) is required to review analytical data and provide a professional evaluation of the data under their PG seal. Contractor can subcontract PG services if a PG is not on contractor staff. PG seal/license verification can be requested by the USG as required.

1.4.7 Inspections of Work Sites: Contractor shall submit to potential federal, state, Army, and installation work site environmental regulatory inspections and/or investigations into noncompliance and fully cooperate with such inspections/investigations by providing the appropriate records and documentation. Environmental regulatory agencies are authorized by law to inspect any work site for environmental compliance with regulatory requirements. If an inspection is conducted, it will not stop or disrupt ongoing contract activities. The inspection will only require the work site environmental officer, or supervisor/manager to answer questions and/or escort the inspector to specific work site areas. Typical environmental work site inspections are conducted in less than 30 minutes with an approximate frequency of one inspection every two months.

1.4.8 Identification of Contractor Employees: All contact personnel attending meetings, answering Government telephones and working in other situations where their contractor status is not obvious to third parties are required to identify themselves as such to avoid creating an impression in the minds of members of the public that they are Government officials. The contractor shall also ensure that all documents or reports produced by contractor personnel are suitably marked as contractor products or that contractor participation is appropriately disclosed. All contractor personnel that require entry to the LEMC/LEAD facility will be required to obtain and wear badges issued by the LEAD security office after a successful background check has been obtained.

1.4.9 Access and general protection/security policy and procedures: Contractor and all associated subcontractor's employees shall provide all information required for background checks to meet installation access requirements to be accomplished by installation Provost Marshal Office, Director of Emergency Services or Security Office. Contractor workforce must comply with all personal identity verification requirements (FAR clause 52.204-9, Personal Identity Verification of Contractor Personnel) as directed by DOD, HQDA and/or local policy. In addition to the changes otherwise authorized by the changes clause of this contract, should the Force Protection Condition (FPCON) at any individual facility or installation change, the Government may require changes in contractor security matters or processes.

1.4.10 For contractors that do not require a CAC but require access to a DoD facility or installation: Contractor and all associated sub-contractors employees shall comply with adjudication standards and procedures using the National Crime Information Center Interstate Identification Index (NCIC-III) and Terrorist Screening Database (TSDB) (Army Directive 2014-05/AR 190-13), applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by government representative), or, at OCONUS locations, in accordance with status of forces agreements and other theater regulations. The contractor shall coordinate with the TPOC for facility access.

2.0 Definitions and Acronyms:

2.1.1 CONTRACTOR. A supplier or vendor awarded a contract to provide specific supplies or service to the government. The term used in this contract refers to the prime.

2.1.2 CONTRACTING OFFICER (KO). A person with authority to enter into, administer, and or terminate contracts, and make related determinations and findings on behalf of the government. Note: The only individual who can legally bind the government.

2.1.3 TECHNICAL POINT OF CONTACT (TPOC). An employee of the U.S. Government designated by the KO to monitor contractor performance. Such appointment shall be in writing and shall state the scope of authority and limitations. This individual has authority to provide technical direction to the Contractor as long as

that direction is within the scope of the contract, does not constitute a change, and has no funding implications. This individual does NOT have authority to change the terms and conditions of the contract.

2.1.4 DEFECTIVE SERVICE. A service output that does not meet the standard of performance associated with the Performance Work Statement.

2.1.5 DELIVERABLE. Anything that can be physically delivered and includes non-manufactured things such as meeting minutes or reports.

2.1.6 KEY PERSONNEL. Contractor personnel that are evaluated in a source selection process and that may be required to be used in the performance of a contract by the Key Personnel listed in the PWS. When key personnel are used as an evaluation factor in best value procurement, an offer can be rejected if it does not have a firm commitment from the persons that are listed in the proposal.

2.1.7 PHYSICAL SECURITY. Actions that prevent the loss or damage of Government property.

2.1.8 QUALITY ASSURANCE. The government procedures to verify that services being performed by the Contractor are performed according to accepted standards.

2.1.9 QUALITY ASSURANCE SURVEILLANCE PLAN (QASP). An organized written document specifying the surveillance methodology to be used for surveillance of contractor performance.

2.1.10 QUALITY CONTROL. All necessary measures taken by the Contractor to assure that the quality of an end product or service shall meet contract requirements.

2.1.11 SUBCONTRACTOR. One that enters into a contract with a prime contractor. The Government does not have privity of contract with the subcontractor.

2.1.12 WORK DAY. The number of hours per day the Contractor provides services in accordance with the contract.

2.1.13 WORK WEEK. Monday through Thursday, unless specified otherwise.

2.2 ACRONYMS:

AFARS	Army Federal Acquisition Regulation Supplement
APS	Application/Permit/Site
AR	Army Regulation
ARMD	Ammonium Perchlorate Rocket Motor Destruction Facility
CAT	Caterpillar Equipment
CFR	Code of Federal Regulations
CFP	Contractor Furnished Property

CONUS	Continental United States (excludes Alaska and Hawaii)
COTS	Commercial-Off-the-Shelf
DA	Department of the Army
DD250	Department of Defense Form 250 (Receiving Report)
DD254	Department of Defense Contract Security Requirement List
DFARS	Defense Federal Acquisition Regulation Supplement
DOD	Department of Defense
FAR	Federal Acquisition Regulation
GFP	Government Furnished Property
KO	Contracting Officer
LEAD	Letterkenny Army Depot
LEMC	Letterkenny Munitions Center
MHE	Material Handling Equipment
OB/OD	Open Burn/ Open Detonation
OCI	Organizational Conflict of Interest
ODC	Other Direct Costs
PADEP	Pennsylvania Department of Environmental Protection
PIPO	Phase In/Phase Out
PLC	Production Logic Controller
POC	Point of Contact
PoP	Period of Performance
PRS	Performance Requirements Summary
PWS	Performance Work Statement
QA	Quality Assurance
QA/QC	Quality Assurance/Quality Control
QAP	Quality Assurance Program
QASP	Quality Assurance Surveillance Plan
QC	Quality Control
QCP	Quality Control Program
RCRA	Resource Conservation and Recovery Act
TE	Technical Exhibit
TPOC	Technical Point of Contact

3.0 Government Furnished Property, Material, Equipment and Services (GFP/M/E/S):

The Government will provide the facilities, equipment, materials, and/or services listed below:

3.1 Facilities: None

3.2 Equipment: None

3.3 Materials: None

3.4 Services: None

3.5 Utilities: None

4.0 Contractor Furnished Property, Materials, Equipment (CFP/M/E):

4.1 General: Except for those items specifically stated to be Government-Furnished in Paragraph 3.0, the contractor shall furnish everything required to perform these services as indicated in Paragraph 1.1.

4.2 Facility Clearance: The contractor shall possess or be eligible to receive and maintain a Letterkenny Army Depot specific clearance from the Defense Industry Security Clearance Office. The contractor's employees, performing work in support of this contract shall have been granted a Letterkenny Army Depot specific clearance from the Defense Industry Security Clearance Office.

4.3 Driver License: All contractor employees shall possess a valid REAL ID driver's license. This will have to be provided upon request to access LEMC.

4.4 Equipment/Tools: The contractor's vehicle must have proper company identification displayed to be allowed on LEMC. The contractor shall have a fire extinguisher inside of vehicle. The contractor shall furnish all equipment and tools required for completion of the requirements of this PWS. The contractor shall also be responsible for the disposal and cleanup of all fluids and materials discarded during the repair process. All the fluids and materials shall be disposed of in accordance with applicable state and federal regulations off site.

5.0 Requirements: What is listed below are summaries and the specific performance work specifications that are to be in compliance with PADEP approved environmental sampling plans. RCRA Subpart X; Permit No. PA6213820503, APS No. 960338, Authorization No. 1215612, OnBase Reference ID 1730 and specifically Subpart X, Environmental Sampling Plans, Appendix E-1 and Appendix E-2. Approved sampling plans: RCRA Subpart X, Appendix E-1 and Appendix E-2 will be provided. The contractor shall Sample/Test in accordance with Environmental Sampling Plans mentioned above. The contractor must base quotation off of PADEP approved Environmental Sampling Plans referenced. The samples listed below are an executive summary and are subject to change and will be incorporated or modified with a contract modification.

5.1 ARMD Sampling Requirements:

The contractor is required to submit the report annually to PADEP and submit copies of all sampling reports to LEMC and LEAD Environmental office. The Contractor shall conduct environmental sampling in accordance with the provisions of the most current version of the Environmental Sampling Plan for Soil, Groundwater, and Surface Water Quality Monitoring for the Ammonium Perchlorate Rocket Motor Destruction (ARMD) Facility.

Soil, surface water, groundwater, stormwater runoff and runoff sediment, will be collected as described in the following section. Samples will be collected from the locations and will be analyzed for the parameters identified in the sampling plan. The

Sampling Plan provides proposed field sample identification and Quality Assurance Quality Control (QA/QC) sample collection for the baseline and annual sampling events. Information on QA/QC samples are included in section 6 of the ARMD Environmental Sampling Plan, Appendix E-2. The following sampling must be done annually and, in the quantities specified within the environmental sampling plan, RCRA Subpart X, Appendix E-2:

5.1.1 Soil Samples: Five (5) annual soil samples (plus QA/QC samples will be collected from the same three locations near the buildings used in the ARMD process, and the same two locations of the central area of ARMD.

5.1.2 Groundwater Samples: Four (4) monitoring wells will be sampled for annual sampling events at the locations depicted on Figure 3 contained in ARMD Environmental Sampling Plan, Appendix E-2. One monitoring well will be located up gradient from the ARMD and three wells will be located down gradient of the ARMD.

Groundwater level measurements will be collected during groundwater sampling events. Water levels from all four wells will be collected on the same day and as close together as possible.

5.1.3 Surface Water Samples: Four (4) surface water samples (plus QA/QC samples will be collected from the four locations in proximity to the ARMD.

5.1.4 Stormwater Runoff and Runoff Sediment Samples:

A total of two (2) stormwater runoff samples and two (2) runoff sediment samples will be collected from two locations. One stormwater runoff sample and one runoff sediment sample will be collected from each location. These locations are labeled in the sampling plan. The lines connected to points A and B on Figure 3 in ARMD Environmental Sampling Plan, Appendix E-2 are storm sewer pipes. There is a small bed of rocks located at the outfall of each pipe. The two stormwater runoff locations will be sampled during annual sampling events. Samples will include both stormwater runoff, if water is present, and runoff sediment during sampling events.

5.1.5 Data Evaluation:

Statistical analysis will compare groundwater data having greater than baseline concentrations once operations begin, as well as comparing down-gradient concentrations with up-gradient concentrations prior to operation, to ascertain if the sampling results are statistically significant. Similarly, statistical analyses will be used to compare soil and sediment data. All environmental monitoring data for surface water, groundwater and soils will be assessed relative to baseline per 40 CFR 264.94 (a) (1). Field verification of significant concentrations may require additional confirmation sampling to verify that the concentration of a specific parameter is valid.

LEAD proposes to use the EPA software ProUCL for analysis of variance (ANOVA) consistent with 40 CFR Part 264.97(h) ANOVA analyses. ProUCL provides numerous

and varied statistical methods and graphical tools to address environmental sampling and statistical issues.

ProUCL can run environmental data sets with and without non-detect (ND) data samples. Calculating upper statistical limits is a primary function of the software and the graphical analyses offered include probability plots, histograms, box plots, and line/trend plots. Results for statistical intervals have several options and relevant cautions. Other available tests that may apply to groundwater monitoring include single and two-sample hypotheses tests, ANOVA, regression, trend evaluation, outlier, and goodness-of-fit tests.

Environmental monitoring data, along with a narrative report and pertinent maps will be submitted to PADEP within 60 days after sample collection. The report to PADEP will include statistical treatment of monitoring data, all sampler field log/notes including weather conditions for day of sampling, monitoring well purge records, contour maps, laboratory data packages with all analytical data (including internal lab QA/QC data), chain of custody sheets and an EDD in a format consistent with the PADEP's LandLinks environmental database.

5.1.5.1 Soil: Analytical data for soil sample results will be compared to the human health and ecological screening levels for soil.

5.1.5.2 Sediment: Analytical data for sediment sample results will be compared to the human health and ecological screening levels for sediment.

5.1.5.3 Downstream Surface Water: Analytical data will be compared to data from the relatively upstream background locations.

5.1.5.4 Groundwater: Analytical data will be compared to data from the background well, monitoring well 1 (MW1).

5.1.6 Report Requirements:

5.1.6.1. Sampler's field notes and well purging records.

5.1.6.2. Complete lab reports including analysis of all samples in accordance with EPA test methods and permitted plans, all QA/QC, and chains of custody. Laboratory must be accredited for the specific analyses by the PA Laboratory Accreditation Program:

Laboratory accreditation information found here :

<https://usg01.safelinks.protection.office365.us/?url=https%3A%2F%2Fwww.pa.gov%2Fservices%2Fdep%2Fapply-for-laboratory-accreditation.html&data=05%7C02%7Cgregory.a.epstein.civ%40army.mil%7Ce191f64666324b063ab608ddae822831%7Cfae6d70f954b481192b60530d6f84c43%7C0%7C0%7C638858596221821700%7CUnknown%7CTWfPbGZsb3d8eyJFbXB0eU1hcGkiOnRydWUslIYiOiwljAuMDAwMCIsIAiOiJXaW4zMilslkFOLjoiTWfPbClslldUljoyfQ%3D%3D%7C0%7C%7C%7C&sdata=bjXAGtbi18Jvsi%2Fv3EDZv4jRzFULqRqh7%2BH62SKH6tY%3D&reserved=0>

5.1.6.3. Data validation and statistical treatment of the data.

5.1.6.4. Groundwater contour map.

5.1.6.5. Narrative evaluation of the data prepared and sealed by a PA licensed professional geologist, which discusses sample collection, data validation, statistical analysis, groundwater flow (rate and direction), and whether the data indicate effects from facility treatment activities.

5.1.6.6. An electronic data deliverable (EDD) spreadsheet in the LandLinks format, acceptable to PADEP environmental monitoring database.

5.2 OBOD (Open Burn/Open Detonation) Sampling Requirements:

The contractor is required to submit the report annually to PADEP and submit copies of all sampling reports to LEMC and LEAD Environmental office. Sediment and Soil Analysis.

5.2.1 Sediment Trap Sampling: One sediment sample from each sediment trap (OB1, OB2, and Flashing Furnace) per year, provided there is sufficient volume of sediment available.

5.2.2 Sedimentation Basin Sampling: One sediment sample of the sedimentation basin dredgings, collected immediately following dredging activities or directly from the sedimentation basin.

5.2.3 Soil Sampling:

5.2.3.1 One soil sample from the base of a freshly fired detonation crater (approximately 10 feet below ground surface [bgs]) in the Long Field.

5.2.3.2 At least one soil sample at ground surface in the footprint of the detonation crater following regrading of the area (Long Field). This soil sample to be taken at the ground surface will be collected within 50 feet of the detonation, per Permit Condition V.A.3.

5.2.3.3 A single surface soil sample at the Short Field on the same day that samples are collected at the Long Field. The sediment and soil samples required in Part V.A.1 though V.A.3 of RCRA Part B Permit No. PA6213820503 shall be analyzed for: all parameters listed in Table 4 of Appendix A of 25 Pa Code §250.1 et seq. (regarding the Department's Land Recycling Program); the inorganic parameters of sulfate, chloride, fluoride, nitrate, and nitrite; and the explosives parameters of nitrocellulose, nitroguanidine, nitroglycerine, perchlorate, 2,4-dinitrotoluene, amino-4,6-dinitrotoluene, 4-amino-2,6-dinitrotoluene, tetryl, 1,3,5-trinitrobenzene, 2,4,6-trinitrotoluene, HMX, and RDX. The following sampling activities must be conducted annually, in accordance with Subpart X, Environmental Sampling Plan, Appendix E-1:

5.2.4 Surface Water Sampling:

Five surface water samples:

5.2.4.1 One from the Back Creek Sub-Basin in the unnamed tributary downstream from OB1 and OB2, on the downstream side of Georgia Avenue.

5.2.4.2 One from the Rocky Spring Branch Sub-Basin in the effluent stream of the sediment pond.

5.2.4.3 One from the old impoundment pond, downstream of Barricade Road.

5.2.4.4 One in the Back Creek sub-basin not affected by OB/OD activities, on the branch of the stream paralleling Georgia Avenue, upstream of the confluence of Back Creek and the tributary that flows from OB2 and unaffected by OB1 drainage.

5.2.4.5 One in the Rocky Spring Branch sub-basin not affected by OB/OD activities. A suitable surface water sampling location exists on the eastern limb of the headwaters of the unnamed tributary to Rocky Spring Branch upgradient of the old impoundment pond, west of Barricade Road and south of Kentucky Avenue extension, and unaffected by drainage from the OD areas and OB2.

5.2.4.6 Surface water sampling should occur within 24 hours of a high rainfall event. A high rainfall event shall be defined as no less than 2 inches of rain per 12-hour period. Surface Water Analysis. The surface water samples required in Permit Condition V.A.4 above shall be analyzed for:

Total and dissolved phases of all parameters required in Permit Condition V.A.6 above; and the parameters of temperature, dissolved oxygen, ammonia nitrogen, specific conductance, total dissolved solids, total suspended solids, pH, alkalinity (as CaCO₃), and hardness (as CaCO₃).

5.2.5 Groundwater Sampling:

Nine monitoring wells: 90-14, 90-16, 90-17 (OB1); 90-3 and 90-5 (OB2); 01-OD2-1, 01-OD2-2, and 01-OD2-3 (OD Area); and 90-18 (Background).

5.2.6 Macroinvertebrate Sampling:

5.2.6.1 Six samples (ODB01 through ODB06) will be collected in the unnamed tributary (referred to herein as the OD stream) between the OD Area impoundment pond and Louisiana Run (a tributary to Rocky Spring Branch).

5.2.6.2 Six samples (OBB01 through OBB06) at an unnamed tributary to Back Creek, west of OB1 and near Georgia Avenue, to assess the stream for impacts from OB1 and OB2. The previous five samples collected from this stream reach were designated as MRB01 through MRB05.

5.2.6.3 The headwaters of Muddy Run were selected on the basis of comparable geology, stream channel morphology and habitat similarity, and stream size. The candidate sample reach is shown on Figure 9 of Environmental Sampling Plan,

Appendix E-1. The benthic macroinvertebrate sampling locations in the Muddy Run reference stream will be selected in the field on the basis of more detailed, current information on substrate characteristics and stream flow obtained during sampling of the compliance monitoring locations in Back Creek and in the unnamed tributary to Rocky Spring Branch.

5.2.6.4 Sampling to occur within same stream reaches that were established in 2001 and initially sampled in January 2002 in order to allow comparisons with prior macrobenthic invertebrate monitoring data.

5.2.7 Resampling Trigger:

5.2.7.1 LEAD Environmental Division, through the TPOC, will make arrangements to resample within 10 days after receipt of surface water or groundwater analytical data that indicates degradation from OB/OD or Flashing Furnace activities.

5.2.7.2 PADEP considers degradation to be a measurable increase in the concentration of one or more contaminants above background in groundwater or surface water.

5.2.8 Data Evaluation:

Following receipt, review and validation of analytical data, a summary report of the OB/OD area and Flashing Furnace sampling will be prepared and submitted to PADEP. The report will include a summary of sampling activities, summary of results, and a narrative data evaluation, including sample location figures, a groundwater contour map and data tables. The data submitted will be prepared by a PA-licensed professional geologist.

Resampling will be triggered by detections of those parameters listed for surface water and groundwater in Table 1 of Environmental Sampling Plan, Appendix E-1, with the exception of iron, manganese, sodium, potassium, zinc, total dissolved solids, hardness, total suspended solids, sulfate, and alkalinity.

5.2.8.1 Soil: Analytical data for soil sample results will be compared to the human health and ecological screening levels for soil.

5.2.8.2 Sediment: Analytical data for sediment sample results will be compared to the human health and ecological screening levels for sediment.

5.2.8.3 Downstream Surface Water: Analytical data will be compared to data from the relative upstream background locations.

5.2.8.4 Groundwater: Analytical data will be compared to data from the background well, 90-18, provided it does not show any effects from OB/OD activities.

5.2.8.5 Benthic Macroinvertebrate and Habitat Assessment: Tabulated in a benthic macroinvertebrate community analysis. A variety of statistical and index methods will be reported including taxa richness, diversity, similarity, and other community indices.

5.2.9 Report Requirements:

5.2.9.1. Sampler's field notes and well purging records.

5.2.9.2. Complete lab reports including analysis of all samples in accordance with EPA test methods and permitted plans, all QA/QC, and chains of custody. Laboratory must be accredited for the specific analyses by the PA Laboratory Accreditation Program:

Laboratory accreditation information found here :

<https://usg01.safelinks.protection.office365.us/?url=https%3A%2F%2Fwww.pa.gov%2Fservices%2Fdep%2Fapply-for-laboratory-accreditation.html&data=05%7C02%7Cgregory.a.epstein.civ%40army.mil%7Ce191f64666324b063ab608ddae822831%7Cfae6d70f954b481192b60530d6f84c43%7C0%7C0%7C638858596221821700%7CUnknown%7CTWfPbGZsb3d8eyJFbXB0eU1hcGkiOnRydWUslIYiOilwLjAuMDAwMCIsIiAiOiJXaW4zMilslkFOljoiTWFpbClslldUljoyfQ%3D%3D%7C0%7C%7C%7C&sdata=bjXAGtbi18Jvsi%2Fv3EDZv4jRzFULqRqh7%2BH62SKH6tY%3D&reserved=0>

5.2.9.3. Data validation and statistical treatment of the data.

5.2.9.4. Groundwater contour map.

5.2.9.5. Narrative evaluation of the data prepared and sealed by a PA licensed professional geologist, which discusses sample collection, data validation, statistical analysis, groundwater flow (rate and direction), and whether the data indicate effects from facility treatment activities.

5.2.9.6. An electronic data deliverable (EDD) spreadsheet in the LandLinks format, acceptable to PADEP environmental monitoring database.

6.0 Documents:

6.1 OCCUPATIONAL HEALTH AND SAFETY (OSHA) STANDARDS

6.1.1 29 CFR, Chapter XVII, Part 1910 – Occupational Safety and Health Regulations

6.1.2 29 CFR 1910 Subpart Z – Toxic and Hazardous Substances

6.1.3 29 CFR 1910.94 Ventilation

6.1.4 29 CFR 1910.1027 Toxic and Hazardous Substances: Cadmium

6.1.5 29 CFR 1910.1026 Toxic and Hazardous Substances: Chromium

6.1.6 29 CFR 1910.1025 Toxic and Hazardous Substances: Lead

6.1.7 29 CFR 1910.146 Permit-Required Confined Spaces

6.1.8 21 CFR Subpart J, General Industry.

6.1.9 OSHA Safety Health Standards

Link: <https://www.osha.gov>

6.2 LEAD Pamphlet 385-4 – Requirements for Security, Safety, and Fire Prevention for Contractors Performing Work on Letterkenny Army Depot (copies may be obtained from Letterkenny Army Depot Fire Department, 717-267-8231)

6.3 LEAD Regulation 385-1 – LEAD Safety and Occupational Health Program. Particularly Chapter 26, which covers LEAD's Lead, Cadmium, and Chromium Program (copies may be obtained from Letterkenny Contracting Office)

6.4 Federal Environmental Laws and Regulations 40 CFR Part 264: Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities. Resource Conservation and Recovery Act (RCRA): Description: Governs the management of hazardous and non-hazardous solid waste, including generation, transportation, treatment, storage, and disposal.

Link: EPA RCRA Overview.

<https://www.epa.gov/rcra/resource-conservation-and-recovery-act-rcra-overview>

RCRA Permits: RCRA Subpart X; RCRA Part B Permit No. PA6213820503, APS No. 960338, Authorization No. 1215612, OnBase Reference ID 1730 – PADEP approved Environmental Plans: Subpart X, Appendix E-1 and E-2

6.5 Clean Air Act (CAA). Description: Establishes the framework for regulating air emissions from stationery and mobile sources to protect air quality.

Link: EPA Clean Air Act Overview:

<https://www.epa.gov/clean-air-act-overview>

6.6 Clean Water Act (CWA). Description: Regulates discharges of pollutants into U.S. waters and sets quality standards for surface waters

Link: EPA Clean Water Act Summary:

<https://www.epa.gov/laws-regulations/summary-clean-water-act>

6.7 Safe Drinking Water Act (SDWA). Description: Protects the quality of drinking water in the U.S., focusing on all waters potentially designed for drinking use.

Link: EPA Safe Drinking Water Act

<https://www.epa.gov/sdwa>

6.8 Toxic Substances Control Act (TSCA). Description: Provides EPA with authority to require reporting, record-keeping, testing, and restrictions relating to chemical substances.

Link: EPA TSCA Summary

<https://www.epa.gov/laws-regulations/summary-toxic-substances-control-act>

6.9 DoD Manual 6055.09-M – DoD Ammunition and Explosives Safety Standards. Description: Establishes explosives safety standards for the DoD to manage risks associated with ammunition and explosives.

Link: DoD 6055.09-M Manual (PDF)

<https://www.denix.osd.mil/ddes/denix-files/sites/32/2022/08/DESR-6055.09-Edition 1-Change-2-251208.pdf>

6.10 Army Regulation (AR) 200-1 – Environmental Protection and Enhancement. Description: Implements federal, state, and local environmental laws and DoD policies for preserving and restoring environmental quality.

Link: AR 2001-1 (PDF)

https://armypubs.army.mil/ProductMaps/PubForm/Details.aspx?PUB_ID=2232

6.11 EPA and Technical Guidance. EPA SW-846 – Test Methods for Evaluating Solid Waste, Physical/Chemical Methods. EPA Method 8330B – Nitroaromatics, Nitramines, and Nitrate Esters by HPLC

Link:

<https://www.epa.gov/esam/epa-method-8330b-sw-846-nitroaromatics-nitramines-and-nitrate-esters-high-performance-liquid>

6.11.1 EPA QA/G-5 – Guidance for Quality Assurance Project Plans (QAPPs)

Links:

<https://www.epa.gov/quality/quality-assurance-project-plan-qapp-guidance>

<https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockkey=20011HPE.TXT>

6.12 National Emission Standards for Hazardous Air Pollutants (NESHAP) and New Source Performance Standards (NSPS)

Links:

<https://www.epa.gov/stionary-sources-air-pollution/national-emission-standards->

[hazardous-air-pollutants-neshap-8](#)

<https://www.epa.gov/stationary-sources-air-pollution/new-source-performance-standards>

6.13 Industry Standards. ASTM International Standards. Description: Provides standardized test methods and practices for environmental sampling and analysis.

ASTM D5298 – Standard Test Method for Measurement of Soil Potential (Suction) Using Filter Paper

Link:

<https://store.astm.org/d5298-16.html>

ASTM D5434 – Standard Guide for Field Logging of Subsurface Explorations of Soil and Rock

Link:

<https://store.astm.org/d5434-03.html>

TECHNICAL EXHIBIT 1

Performance Requirements Summary

This Performance Requirements Summary includes performance standards. The Government will use these standards to determine contractor performance and shall compare contractor performance to the Acceptable Quality Level (AQL).

PWS Paragraph	Task	Performance Standard	Acceptable Quality Levels (AQL)	Surveillance Method / By Whom
5.1	The contractor shall perform Water, soil, and Macroinvertebrate analysis and provide reports on ARMD in compliance with PADEP approved Environmental Sampling Plan	The contractor shall coordinate sample collection, perform water and soil sampling, and provide test results to PADEP/LEAD Environmental/LEMC	0 (zero) failures to supply report within 20 calendar days of sampling.	100% inspection – TPOC will inspect.
5.2	The contractor shall perform Water, soil, and Macroinvertebrate analysis and provide reports on BO/BD in compliance with PADEP approved Environmental Sampling Plan	The contractor shall coordinate sample collection, perform water and soil sampling, and provide test results to PADEP/LEAD Environmental/LEMC	0 (zero) failures to supply report within 20 calendar days of sampling.	100% inspection – TPOC will inspect.

TECHNICAL EXHIBIT 2

Deliverables Schedule

Deliverable	Frequency	Number of Copies	Medium/Format	Submit To
1.4.6.1 TARP Training	30 calendar days following award and annually afterwards	1 per employee	Provide certificates in as electronic submission via email	LEMC TPOC
1.4.6.2 - Level I OPSEC Awareness training	30 calendar days following award and annually afterwards	1 per employee	Provide certificates in as electronic submission via email	LEMC TPOC
5.1 Test Results and analysis of ARMD	Within Ten calendar days after completion of samples testing	1	Electronic	PADEP/LEAD Environmental/LEMC
5.2 Test Results and analysis of OBOD	Within Ten calendar days after completion of samples testing	1	Electronic	PADEP/LEAD Environmental/LEMC

Attachments to PWS:

1. Appendix E-1 OBOD Sampling Plan
2. Appendix E-2 ARMD Sampling Plan